DEPARTMENT OF COMPUTER SCIENCE
Capital University of Science and Technology, Islamabad

Date: 12/10/2026 **CS4613 – Machine Learning** **Marks: 10**

Assignment No: 01 **Instructor: Engr. Dr. Jalees Ur Rahman** **Deadline: 18/10/2026**

Before performing the assignment, keep in mind the following rules:

1. **Internet and book usage are allowed, but you should know what is written in the solution.**
2. **You must provide the solutions as Jupyter Notebook (IPYNB) files.**
3. **Perform all the questions.**
4. **Late submissions are not allowed.**
5. **Avoiding any of the above rules will lead to a deduction.**

Learning Outcomes:

1. **Describe the basics of ML and its flow**
2. **Implementation of different algorithms**

Question 1: ML Flow (5)

Write a Python program in a Jupyter Notebook that explains all steps of the ML flow in a Markdown cell. Then write the code cells for:

- Download and load any regression or classification dataset from the internet.
- Preprocess the dataset by doing some feature engineering and feature selection.
- Train any ML model suitable for the dataset you have chosen.
- Test and validate the model and show predictions for an unseen sample.

Question 2: Candidate Elimination Algorithm (5)

Write a Python program in a Jupyter Notebook that explains why the Find-S algorithm is not useful compared to Find-S in a Markdown cell. Then write the code cells for:

- Create at least 50 samples of any binary classification problem with at least 5 attributes.
- Pass this data through the Candidate Elimination Algorithm function you wrote and display all the rules that fit the data.

Explain in a Markdown cell:

- Is the model converged or not?
- Why is Candidate Elimination a better choice than Find-S for your data?